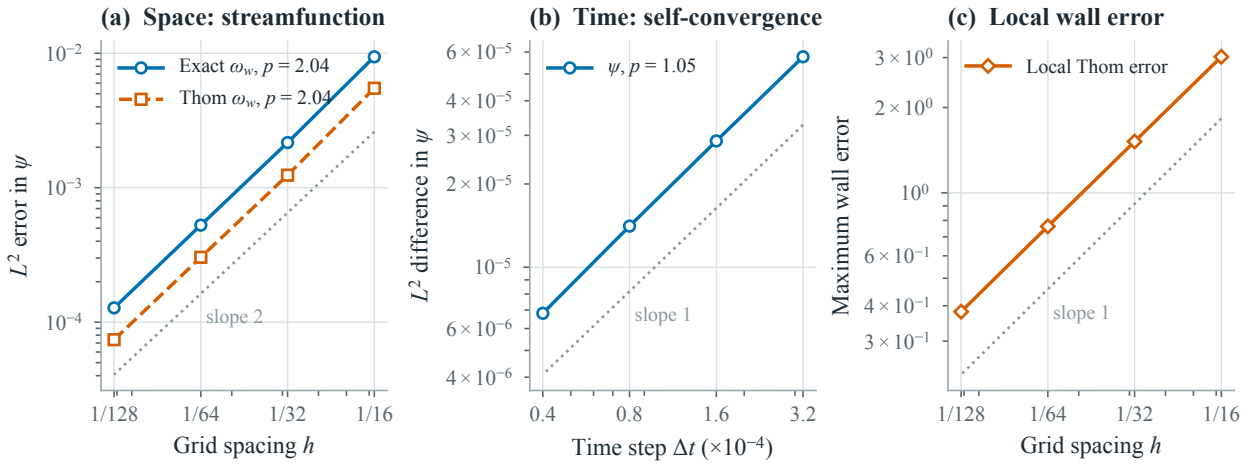

(a) Space: streamfunction
Exact ω_w, $p = 2.04$
Thom ω_w, $p = 2.04$
slope 2
L^2 error in ψ
10^{-2}
10^{-3}
10^{-4}
1/128
1/64
1/32
1/16
Grid spacing h
(b) Time: self-convergence
ψ, $p = 1.05$
slope 1
L^2 difference in ψ
6×10^{-5}
4×10^{-5}
3×10^{-5}
2×10^{-5}
10^{-5}
6×10^{-6}
4×10^{-6}
0.4
0.8
1.6
3.2
Time step Δt ($\times 10^{-4}$)
(c) Local wall error
Local Thom error
slope 1
Maximum wall error
3×10^{0}
2×10^{0}
10^{0}
6×10^{-1}
4×10^{-1}
3×10^{-1}
1/128
1/64
1/32
1/16
Grid spacing h